

A Smart Control Automation Framework for the Production of Hydrogen Energy using Fuel Cells

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Abstract

The paradigm of smart manufacturing or digital automation is the evolving descriptor to efficiently operate chemical process plants that focus not only on the economic and performance objectives but also on the sustainability. Leveraging the technology of multi-parametric model predictive control also known as ‘MPC-on-a-chip’ or explicit MPC, we propose a new framework for the smart control automation of process systems to seamlessly integrate the regulatory control layer with the supervisory control layer of the traditional industrial automation hierarchy. The novelty of this framework is that it not only effectively integrates the automation hardware with the control software but also brings operational intelligence by introducing high fidelity modelling carried out at the multiple scales that captures the complex behaviour of the process plant in real-time. This is a step change from the current industrial practice where detailed modelling activities are carried out to design plants and this meaningful knowledge is either lost or ignored while carrying out the activities of process automation, which otherwise relies entirely on the data driven models after the plant is fully built. This paper presents a step by step procedure of the proposed smart control automation framework to

efficiently operate fuel cell power plant. Although the work focuses on the production of hydrogen energy using polymer electrolyte membrane fuel cells, the proposed framework can be trivially extended to all the other type of fuel cells as well as other conventional and renewable energy generation systems.

Keywords:

Fuel cell, model predictive control, smart manufacturing, digital automation

1. Introduction

Smart manufacturing is the intensified application of ‘manufacturing intelligence throughout the manufacturing and supply chain enterprise that is facilitated by the pervasive use of advanced sensor-based data analytics, modeling, and simulation, [1]. Control system is critical for bringing about manufacturing intelligence to operate modern process plants. It is typically implemented using a combination of hardware that include a network of field devices, workstations, intelligent microcontrollers and software platform that deploy the process control strategies. The microcontroller based sensors and actuators are becoming ubiquitous in manufacturing operations that generate a very large (volume, velocity, or variety) sets of data. A recent study by Hedberg et al., [2] observed that the manufacturing “lifecycle is drowning in data, but starving for information” and proposed a data driven solution based on information technology advances (e.g., data analytics, service-oriented architectures, and networking) that is intelligent, self-learning, and self-aware. This data-centric solution can create a digital cloud-based manufacturing revolution, [3], that integrate operational technology (e.g., hardware and software for sensing, monitoring, and control of product and processes) for

reducing costs, improving productivity, and increasing output quality. These approaches allow inter-operability of data but not the interaction of underlined technologies. Towards that end, a smart manufacturing solution can benefit from the concepts of smart process plants, [4], to unravel scientific phenomena, hidden patterns, correlations and other meaningful operational insights that also guarantee economic value of the control strategies (see chapter 13, [4]). White, [5], identified the incentives for the cumulative and combinatorial aspects of smart microcontroller based field devices and made a compelling case for predictive control versus traditional reactionary type of control system. In this work, we focus on achieving the digital manufacturing revolution by deriving the smart control automation strategies using multi-parametric programming, [6], and control, [7], to operate complex chemical process plants. In a typical industrial setting, field devices are connected

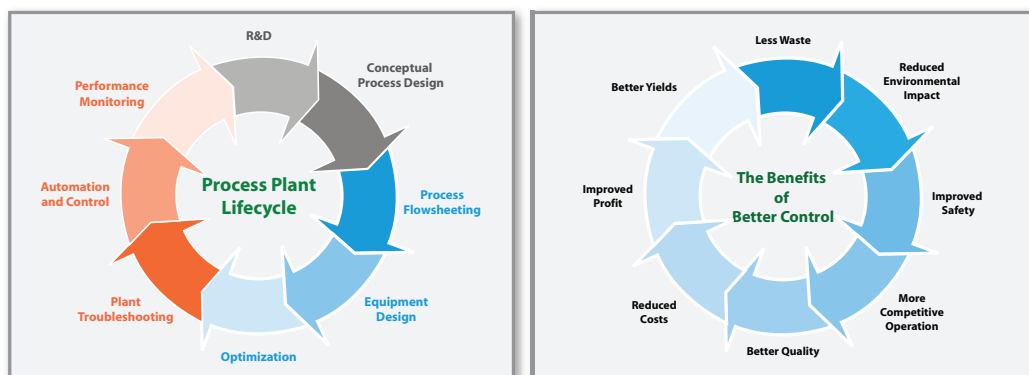


Figure 1: The process plant life cycle and the benefits of better control

to a centralized control system called distributed control system on which various operational strategies are deployed. The activities involved therein with regard to the design and operation are shown in Figure 1. Figure also

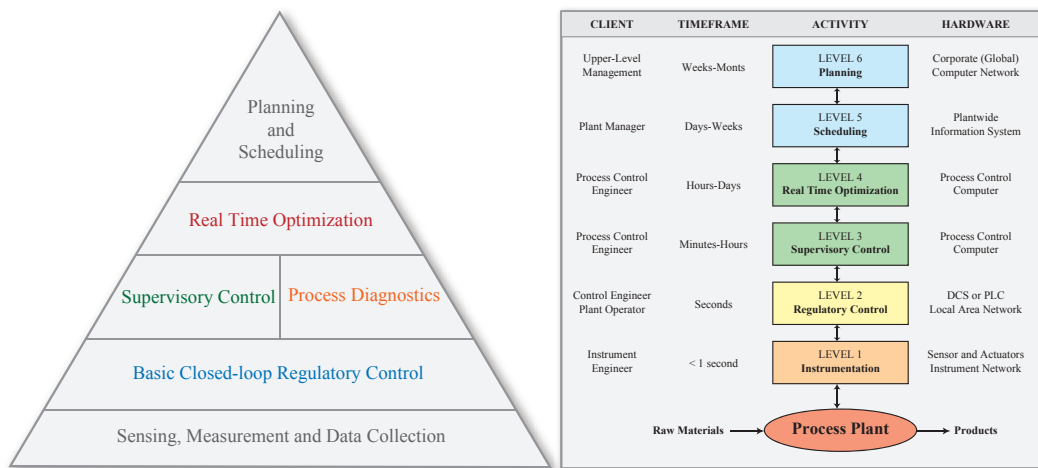


Figure 2: Typical industrial control automation hierarchy

highlights the benefits of well-designed and well-maintained control strategy. In order to achieve these benefits, Figure 2 depicts the hierarchical decomposition of operational management activities and various control functions taking place in a typical process industry. The classical regulatory control system of proportional-integral-derivative that is P only or PI or PID operate at Level 2. In the supervisory form Advanced Process Control (APC) provide the setpoints for these basic regulatory control loops. Thus, APC and the lower level loops are integral part of the overall control structure and must work together to achieve the desired control objectives of minimizing deviation of process variables from their respective setpoints. Currently, there exists a gap in the seamless integration of basic regulatory process control loops/layer with management/supervisory level optimization and control in operating chemical process plants. In addition, process modelling that captures the complex behaviour of process plant has not been fully integrated in

real time in the industrial automation hierarchy. Typical industrial practice is to use modelling to design plants and then carry out activities of process automation after the plant is fully built.

Control technology is shifting gears and so is the role of APC. Moving away from the regulation only type of issues, APC is now being considered as the engine that drives the dynamic economics of process control. The interaction of the Real Time Optimizer (RTO) at Level 4 and the APC at Level 3 is important so as to incorporate these meaningful economics into the control architecture. The integration of RTO that further improves the performance of regulatory control loops was first proposed by [8] and later improved by [9] as a two layered approach. Integration of such strategy into a single layer called economic MPC was proposed by [10]. On the other hand the deployment of off-line or explicit MPC via multi-parametric optimisation techniques by [11] addressed the issues of how the APC could be administered to the applications having fast dynamics (e.g, fuel cells, automotive and biomedical applications) or systems to which on-line implementation of APC is expensive or not possible. In this work, we use these explicit MPC strategies to presents a smart control automation framework to efficiently operate the fuel cell energy system. We discuss the proposed framework from the context of polymer electrolyte membrane fuel cell. Our framework not only integrates the basic layers of automation hierarchy but also brings operational intelligence by integrating aspects of high fidelity modelling while making operating decisions. Additionally, we propose a hardware integration framework for the seamless integration of process process operation activities taking place with the data from the process plant as shown in Figure 3.

Thus, this work sets the foundation for implementing the new paradigm of smart control automation to the process industry.

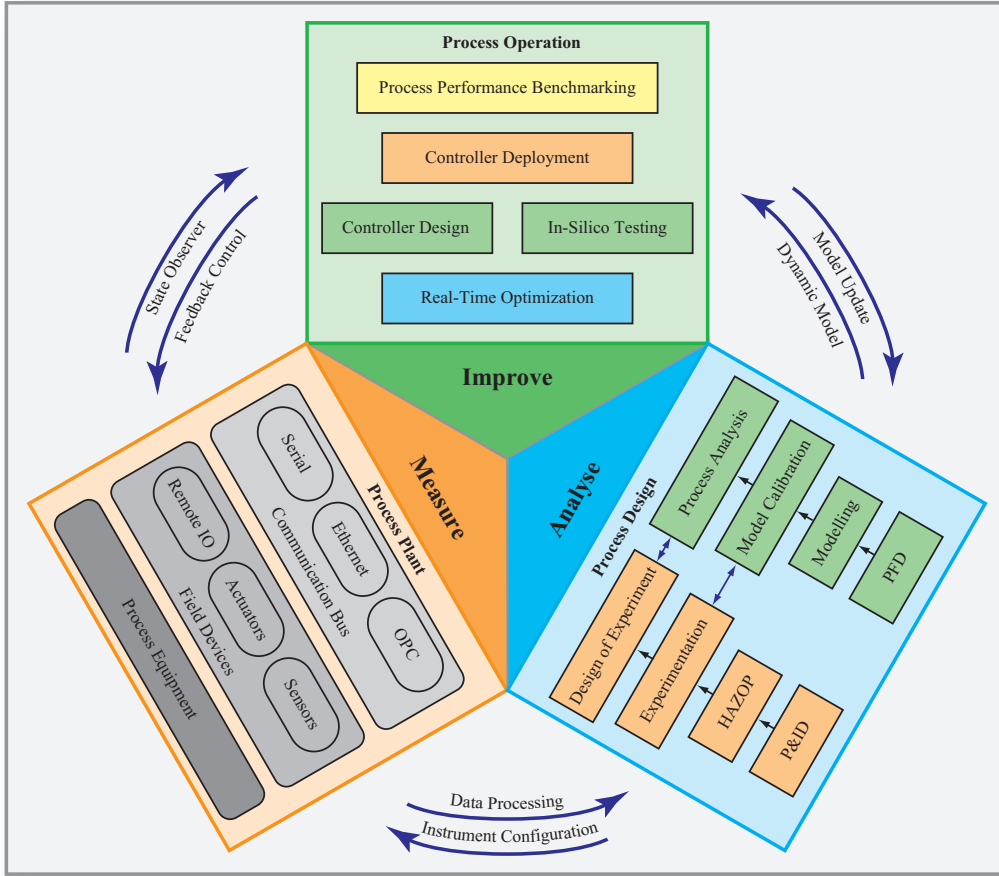


Figure 3: Seamless integration framework for smart process automation

Smart automation is expected to revolutionize process operation by bringing manufacturing intelligence at every level to reduce operating costs, raw material and energy usage while at the same time by maximize product quality, operational performance and safety. Smart automation require a physical infrastructure to collect and analyze data. As a part of this research pro-

gram, a state-of-the-art experimental test platform for polymer electrolyte membrane fuel cell is designed and built in-house to experimentally validate the mathematical models and deploy ‘MPC-on-a-chip’ advanced control strategies. This fuel cell power plant thus becomes a prototype testbed for demonstrating some of these industry relevant concepts. In the next section, we state the motivation for the control of fuel cell system and in the following section describe the proposed framework.

2. Operation of Fuel Cell System

With hydrogen envisioned as an energy carrier, fuel cell is the most efficient and sustainable energy enabling technology. Amongst all the available fuel cell types, a polymer electrolyte membrane fuel cell is the most widely researched fuel cell type. Operating at a low temperature and having higher energy conversion efficiency compared to internal combustion engines, it offers distinct benefits and has found widespread adoption in many areas including the transport and stationary energy sectors. Regardless of the application and the type of fuel cell used, integration of balance of plant system brings additional challenges affecting durability and performance of the overall system, [12]. Although the fuel cell stack technology has reached a level of maturity, the systemwide control automation for better electricity, water and thermal management still remains the main research priorities. The majority of fuel cell system failures and forced outages ($\sim 90\%$) are due to a lack of system integration and are the result of balance of plant events, [13]. These statistics reinforce the need for better system integration and more efficient controller design. The three important aspects for improving fuel cell sys-

tems are: (a) reducing manufacturing and operational costs; (b) improving safety, durability and reliability; (c) improving operational performance by efficient controller design. These three aspects are intertwined and the effective modelling, optimisation and smart control automation strategy can play a key role in addressing operational challenges and significantly improving system performance, durability and operational costs.

The real-time control of fuel cell system using a well established technology of model predictive control requires an accurate mathematical model of the system. The dynamic phenomenon taking place inside fuel cell involves complex interaction of mass transfer, energy transport and electrochemical kinetics. These uncertain dynamics determine the overall operational performance of the system. In addition, optimal thermal, water and electricity management allows for effective use of the fuel cell system's byproduct (mainly heat) leading to the substantial increases in overall system efficiency. However, the traditional real-time model predictive control requires computing machinery to run the underlined optimisation algorithms. Its traditional implementation on a dedicated network of workstation compounds to increase in manufacturing and operational costs. This issue can be resolved by multiparametric controller embedded on a low cost 'system-on-chip' hardware. Unfortunately, due to the fast dynamics of the fuel cell system, controller designs thus far are restricted to the classical PID type of regulatory controllers only. Our emphasis is on the implementation of more advanced model predictive controller (MPC) that can effectively control the fast uncertain system dynamics while guaranteeing feasibility and stability of the system. Since the success of MPC heavily relies on the underlined dynamic

model of the system, the development of mathematical model suitable for control implementation is a key step. Therefore, the primary focus is to develop a mathematical model of fuel cell system and its auxiliary subsystem. The secondary goal is to investigate new multiparametric control algorithms in operating fuel cell system.

3. Smart Control Automation Framework

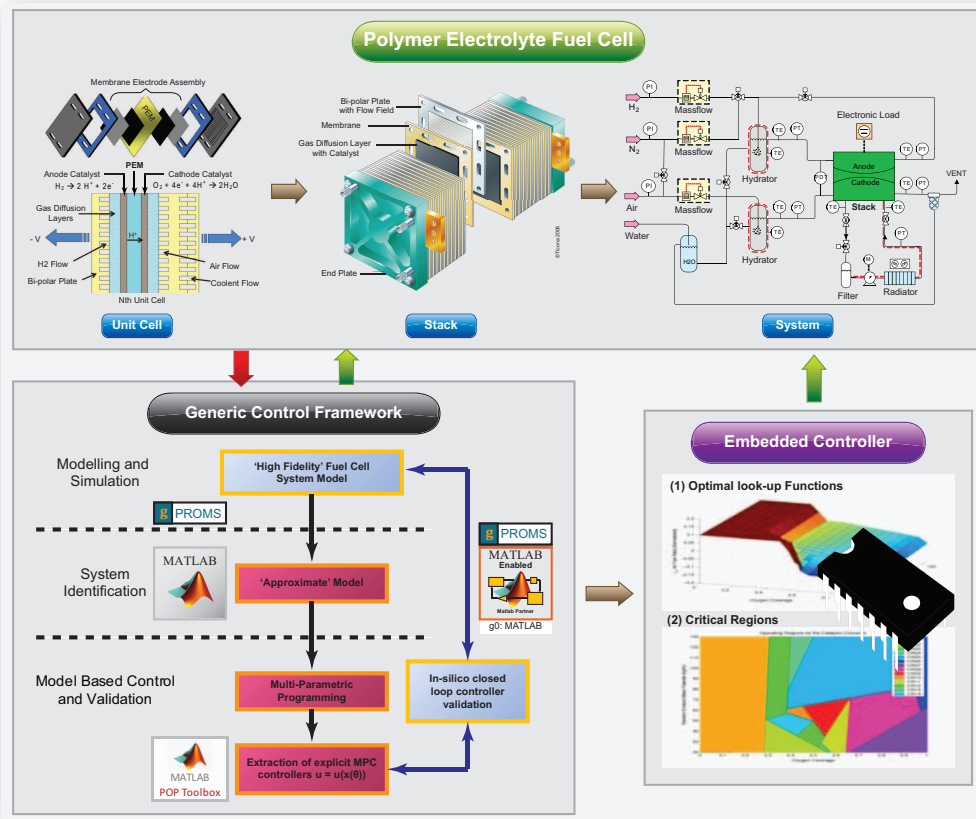


Figure 4: Framework for Fuel Cell Modelling and Control Development

A framework for the real-time control of fuel cell energy system is outlined in Figure 4. It is characterized by the combination of hardware and software functions. This comprehensive smart control automation framework features integration of three elements: (1) fuel cell plant operating at three different scales namely unit cell, stack and integrated plant; (2) the software platform that processes data, performs data analytics and computes optimal control laws; and (3) microcontroller platform that controls the plant. The second element interacts with the plant. It collects data to validate the high fidelity model. A model for the unit cell is first developed and validated against the experimental data. This model is used to build a fuel cell stack model. Next, we design multiparametric model predictive controller using a linearized model of the plant. The performance of this controller is first tested by conducting a number of in silico experiments. Finally, the controller is deployed on a microcontroller platform.

Each of the three elements and activities therein discussed above are generic in nature. For example the modelling step is generic and different type of fuel cells models can be easily plugged into the framework. Similarly, any of the available explicit MPC algorithms can be used to design controller. Since the resulting control law is an affine map of initial operating conditions of the plant, the controller deployment step is straightforward. The following sequel presents the details of the software platform that is developed to control fuel cell system. This platform has three key components: (a) model of the fuel cell, (b) approximate model and (c) control algorithms. These activities operate between Level 4 logistics/planning functions and Level 2 manual/automated process control functions shown in Figure 2. These ac-

tivities are consistent with the Level 3 manufacturing operations and control definitions given in IEC 62264-1 enterprise-control system integration standards.

The section below presents a detailed high fidelity model of the fuel cell system. The succeeding sections describe the model approximation and controller design steps that guarantees safety, stability, durability, performance and improved economics for the fuel cell system operating under the influence of varying supply-demand scenarios and variations in critical parameters.

4. Fuel Cell Model Development

Control oriented model is a key to achieving desired thermal response while effectively managing electricity and water generated. With reasonable computational complexity, lumped control oriented model not only help understand the nature of complex phenomena taking place but also help maintain transient behaviour under load variations with closed-loop feedback. The control oriented modelling work was carried out by number of authors (see review by, [14]). Motivated by the fuel cell model as a stirred tank reactor proposed by, [15] and [16], this work presents a model with the electrochemical relationships that are validated using experimental data from our unit cell experiments, [17].

4.1. Model Assumptions

We assume that all the gases are ideal. The fuel cell stack is fed with pure hydrogen at anode and it reacts quickly and completely. At cathode oxygen is taken from the air supplied by the fan with atmospheric air molar composition $O_2:N_2 = 21:79$. The cathode exhaust contains water as vapor,

excess oxygen and unreacted nitrogen as inert gas. At cathode temperature is stable and constant at all time due to uniform mixing. We further assume that the lower heating value (LHV) of the fuel input.

4.2. Fuel Cell Electrochemistry

During the operation, cell potential undergoes various overpotential losses which are is given by,

$$E_{cell} = E_{ner} - E_{act} - E_{ohm} - E_{con} \quad (1)$$

These losses are explained below.

4.2.1. Nernst Overpotential

The free energy of formation of water is given by

$$\Delta G_{f_{H_2O}} = \Delta G_{f_{H_2O(g)}}^o + R T \ln \left(\frac{a_{H_2O}}{a_{H_2} \sqrt{a_{O_2}}} \right) \quad (2)$$

where ΔG_f and ΔG_f^o are the free energies available for electrical work at operating and standard conditions ; a_{H_2} , a_{O_2} and a_{H_2O} are the activities of species involved which are for ideal gas equal to the concentration of the species; R is the gas constant; T is the operating temperature. Note that we are using the lower value since water formed is assumed to be in vapor form. With free energy change $\Delta G = -nFE$ where E is the reversible potential and F is Faraday's constant, equation (2) leads to following Nernst equation with E^o as the open circuit potential,

$$E_{ner} = E^o - \frac{RT}{nF} \ln \left(\frac{C_{H_2O}}{C_{H_2} \sqrt{C_{O_2}}} \right) \quad (3)$$

4.2.2. Activation Overpotential

At low current densities the activation loss is experienced due to the slow kinetics of electrochemical reaction creating barrier to the flow of electrons and reversal transfer of charge. The activation polarization is represented by,

$$E_{act} = \frac{RT}{\alpha n F} \ln \left(\frac{j + j_{loss}}{j_o} \right) \quad (4)$$

where loss in current j_{loss} is assumed to be zero and j_o is the exchange current density. The temperature dependent exchange current density j_o is given by,

$$j_o = n F k_r e^{(-\beta n F E^o / RT)} \quad (5)$$

where k_r is the associated with the speed of reaction.

4.2.3. Ohmic Overpotential

The next loss in potential is due to the internal electronic resistance through electrodes and external circuit as well as the ionic resistance in electrolyte. It varies proportionally to the increase in current given by Ohm's law,

$$E_{ohm} = \gamma j \quad (6)$$

where the internal ionic resistance γ of the cell is due to the interplay of temperature and humidity. During the experimentation when the cell was operated at higher current densities we observed that the internal ionic resistance is also a function of current density where fuel cell performance starts to deteriorate due to increase in internal resistance. At higher current densities, cell begins to dry out and proton finds it difficult to pass through the membrane. Internal ionic resistance is an important parameter that dictates

ionic crossover impacting the overall cell performance. Exact measurement of phenomena describing transient nature of internal resistances is a challenging task and is part of our future work. However, for the safe operating regime the bulk measurement of this quantity depends on the cell thickness which can be described by following relationship.

$$\gamma = \frac{t_m}{\sigma} \quad (7)$$

where t_m is the membrane thickness and σ is the ionic conductivity. In this work we propose the following temperature and humidity dependent relationship of ionic conductivity,

$$\sigma = \psi e^{\left(\tau - \frac{\theta}{T}\right)} \quad (8)$$

where τ is the exponent of ionic conductivity, θ is exchange ionic conductivity and ψ membrane moisture content. We consider ψ follows an empirical relationship described by a cubic polynomial in relative humidity ϕ ,

$$\psi = a_0 + a_1 \phi + a_2 \phi^2 + a_3 \phi^3 \quad (9)$$

4.2.4. Concentration Overpotential

The drop in potential due to mass transfer effect also known as concentration loss occurs when reactant species finds difficulty in reaching active catalyst sites. It occurs at all operating conditions but is prominent at higher current densities. This polarization limits the amount of current that can be passed through the cell. This diffusion limited overpotential relationship is given by,

$$E_{con} = \frac{\zeta RT}{nF} \ln \left(\frac{j_L}{j_L - j} \right) \quad (10)$$

where γ is mass transfer activity coefficient determining limiting current density and j_L is the limiting current density. In practice, this effect is also approximated by following empirical relation when limiting current density is not exactly known or cell is operated well below limiting value,

$$E_{con} = \frac{\zeta RT}{nF} e^{nj} \quad (11)$$

4.3. Validation of Electrochemistry

From the experimental data reported in [17], two operating scenarios namely normal and abnormal operation are chosen to validate the electrochemical model. Fuel cell is considered as operating normally when the relative humidity is 100% and the oxygen stoichiometry is 2.5 times actually required. It is considered as operating in abnormal condition when the relative humidity drops to 50% and oxygen stoichiometry to its exact required value. Figures 5, 6 and 7 presents the polarization characteristics of the cell. Table 1 below gives the values of identified parameters for the chosen operating scenarios. These two scenarios gives the upper and lower limits of the parameters. The fuel cell operates in the envelop of these two operating scenarios.

5. Dynamics of Fuel Cell

The fuel cell stack is built by connecting number of unit cells in series. A simplified schematic illustrating inner workings of a unit cell is shown in Figure 8. The unit cell is made up of a polymer electrolyte membrane sandwiched between anode and cathode electrodes while each of the gas diffusion layer and the flow chamber is represented by a continuous stirred

Table 1: Experimentally validated model parameters

Parameter	Normal	Abnormal
α	0.9	0.6
β	0.11	0.125
γ	2.4×10^{-4}	2.7×10^{-4}
κ	0.2	0.3
ϕ	1.0	0.5
ζ	0.85	0.95
λ_{O_2}	2.5	1.0
λ_{H_2}	1.5	1.5
j_L	1510	1120

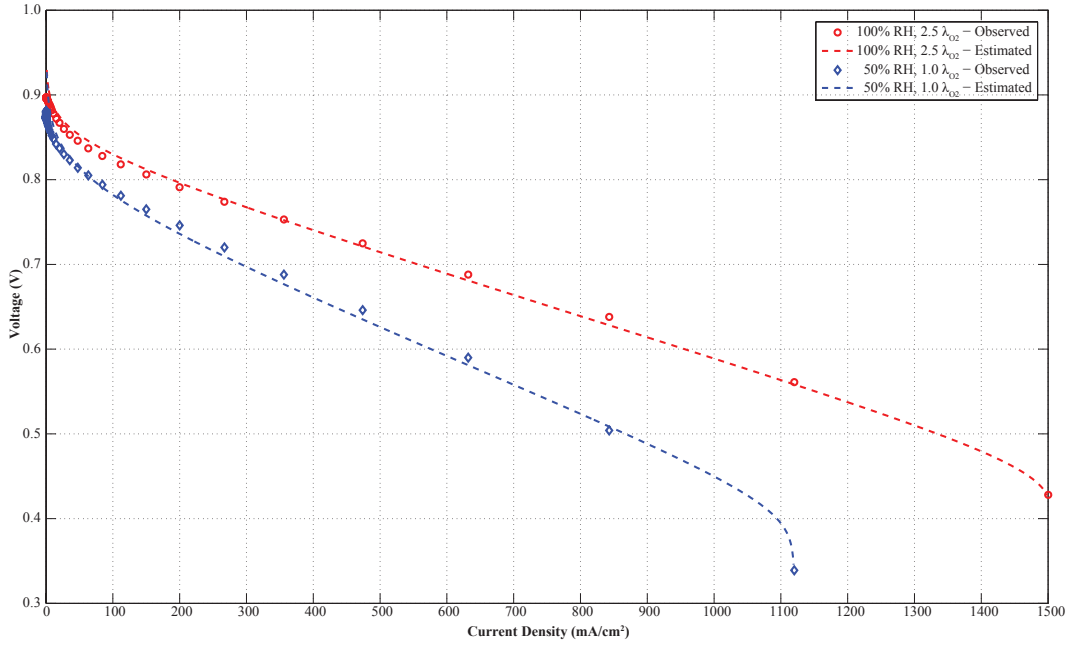


Figure 5: Model Validation of Electrochemical Polarization

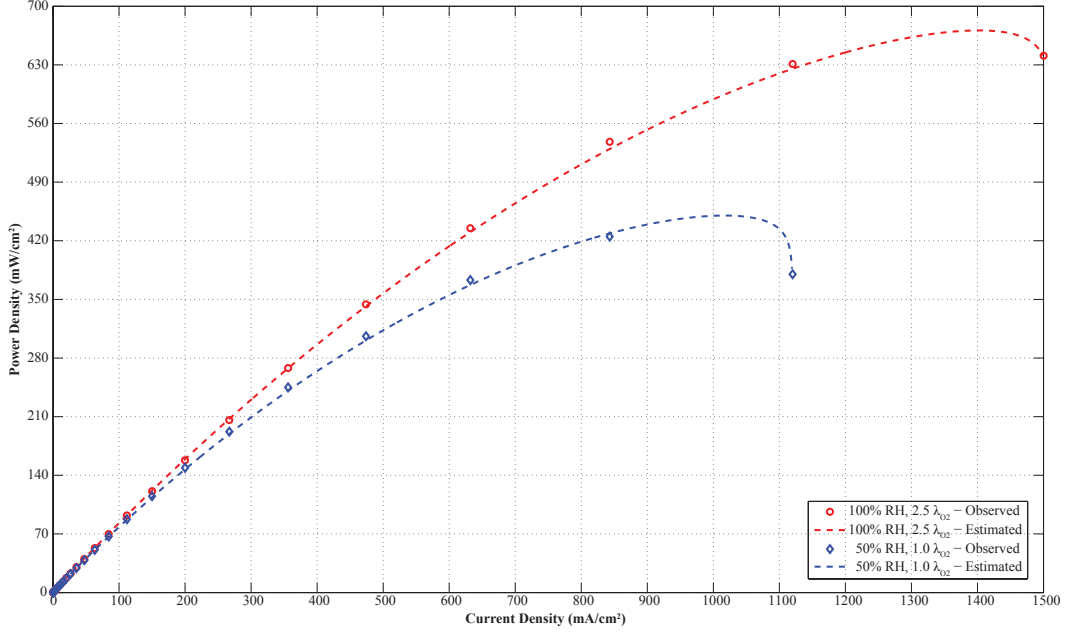
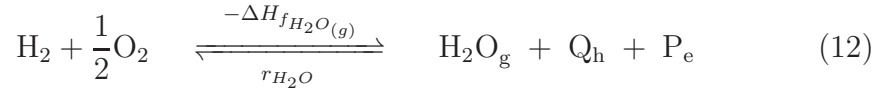


Figure 6: Model Validation of Electrochemical Polarization

tank reactor. The temperature is controlled by cooling fluid typically water circulating in the cooling jacket. Heat is released by two mechanisms—reaction and ohmic heating. Reaction takes place inside the cathode chamber while ohmic heat is generated in the solid block due to flow of electron. The solid block exchanges this heat with the cathode chamber, the surrounding, and the cooling jacket.

In this section we develop the dynamic model of fuel cell using the overall reaction,



The rate of generation of water vapor per unit area of the membrane is given

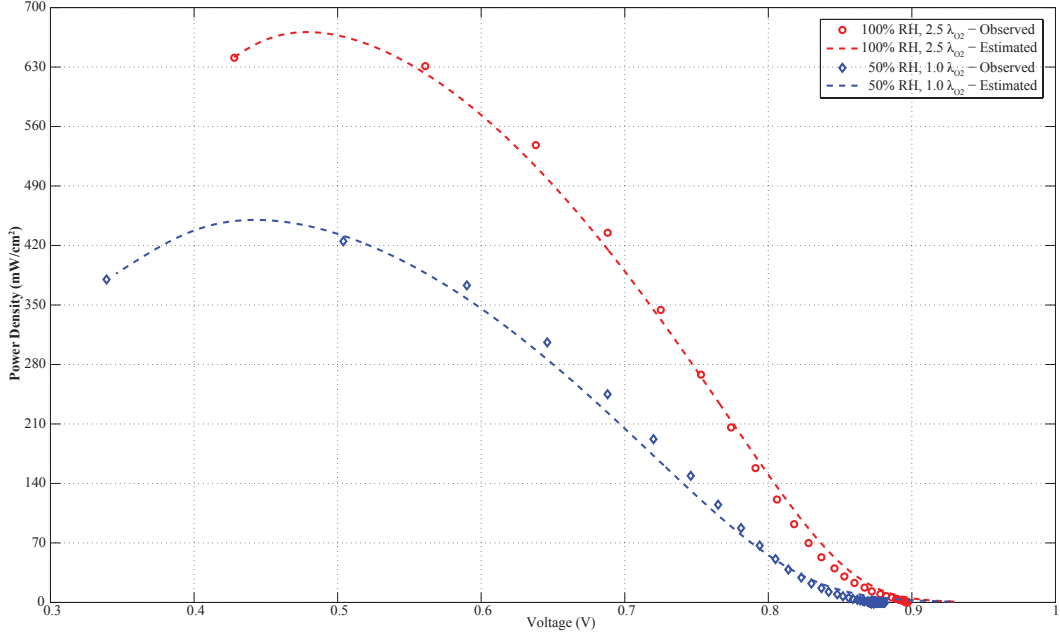


Figure 7: Model Validation of Electrochemical Polarization

by,

$$r_{H_2O} = \frac{j}{nF} \quad (13)$$

where j is the current density, F is the Faraday's constant, and n is the number of electrons taking part in the electrochemical reaction.

5.1. Mole Balances

Due to assumptions made we neglect the material balance on the anode side and only focus our attention on the cathode chamber considering overall reaction. With the notations described in the nomenclature section 9, the

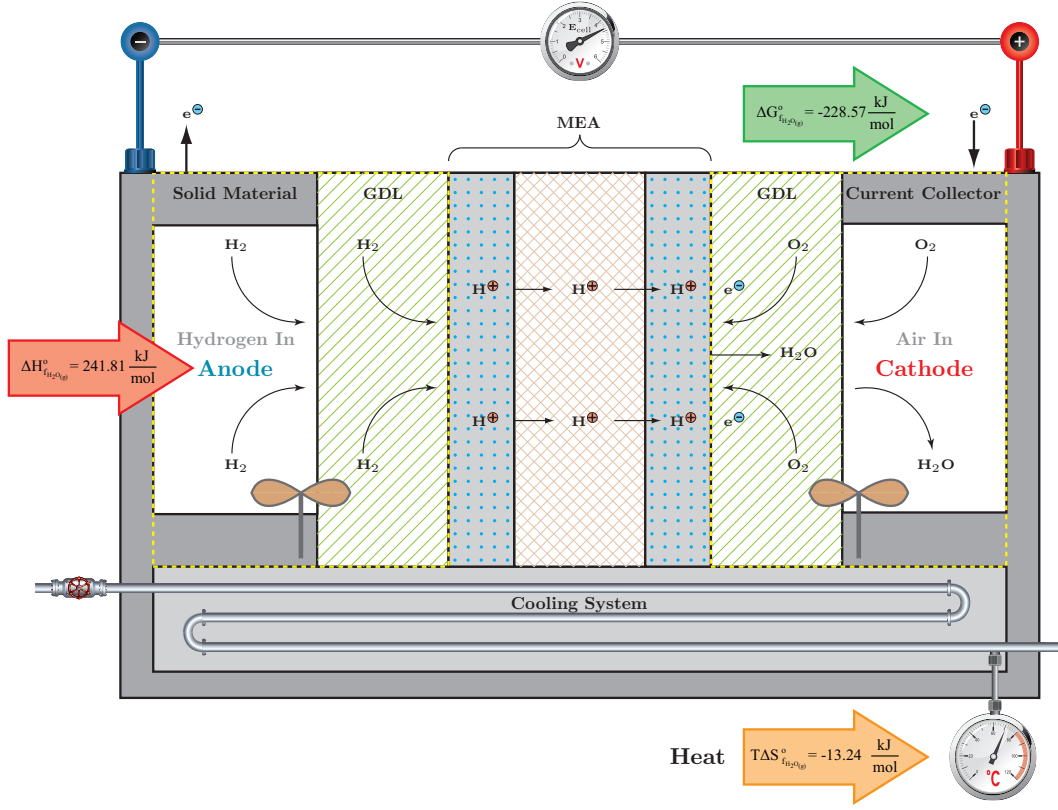


Figure 8: Schematic illustrating the inner working principle of water cooled fuel cell

species mole balances results into following equations,

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_c^{in} - C_{O_2} F_c^{out} - \frac{1}{2} r_{H_2O} A_m \quad (14)$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} F_c^{in} - C_{H_2O} F_c^{out} + r_{H_2O} A_m \quad (15)$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_c^{in} - C_{N_2} F_c^{out} \quad (16)$$

5.2. Overall Mass Balance at Cathode

Assuming no accumulation of overall mass, the expression for cathodic outflow is given by,

$$(C_{O_2} + C_{N_2} + C_{H_2O}) F_c^{out} = (C_{O_2}^{in} + C_{N_2}^{in} + C_{H_2O}^{in}) F_c^{in} + \frac{1}{2} r_{H_2O} A_m \quad (17)$$

5.3. Energy Balance at Cathode

Due to assumption of uniform mixing, non-isothermal energy balance results into following equations with nomenclature given in section 9,

$$V_c \frac{dT}{dt} = F_c^{in} T_c^{in} - F_c^{out} T + \frac{Q_h A_m}{\rho_a C_{p_a}} - \frac{U_c A_c}{\rho_a C_{p_a}} (T - T_s) \quad (18)$$

$$V_s \frac{dT_s}{dt} = \frac{Q_e}{\rho_s C_{p_s}} + \frac{U_c A_c}{\rho_s C_{p_s}} (T - T_s) - \frac{U_j A_j}{\rho_s C_{p_s}} (T_s - T_j) - \frac{U_e A_e}{\rho_s C_{p_s}} (T_s - T_a) \quad (19)$$

$$V_j \frac{dT_j}{dt} = F_j (T_j^{in} - T_j) + \frac{U_j A_j}{\rho_j C_{p_j}} (T_s - T_j) \quad (20)$$

The energy generated in the form of heat Q_h is given by,

$$Q_h = -\Delta H_{f_{H_2O(g)}} r_{H_2O} - P_e + Q_e \quad (21)$$

The ohmic heating of the cell due to joules law electrical heating, Q_e , and temperature dependent heat of formation of water given by,

$$Q_e = \gamma j^2 \quad (22)$$

$$\Delta H_{f_{H_2O(g)}} = \Delta H_{f_{H_2O(g)}}^o + \int_{T_o}^T C_p dT \quad (23)$$

Substituting for heat capacities of species we get,

$$\Delta H_{f_{H_2O(g)}} = \Delta H_{f_{H_2O(g)}}^o + \Delta C_p (T - T^o) \quad (24)$$

where

$$\Delta C_p = C_{p_{H_2O}} - C_{p_{H_2}} - \frac{1}{2} C_{p_{O_2}} \quad (25)$$

Finally, the electrical power generated at steady state is simply $P_e = jE_{cell}$ or the dynamic accumulation of power with cell efficiency η is,

$$\frac{dP_e}{dt} = j E_{cell} - \frac{1}{\eta} P_e \quad (26)$$

5.4. Fuel Cell Stack Model

Stack is made up of number of unit cells, N_c , connected in series. Therefore the only changes that are required in the model derived for unit cell are to multiply the rate expression and power produced by number of cells as follows,

$$r_{H_2O} = \frac{j}{nF} N_c \quad (27)$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} P_e \quad (28)$$

5.5. Simulation of Dynamic Model

Using the experimentally validated parameters of the electrochemical model given in section 4.3, the dynamic response of this model with initial conditions: oxygen concentration of 0.21, water concentration of 0.1, nitrogen concentration of 0.79, cell temperature of 80 °C, solid temperature of 80 °C, and initial power of zero watt is presented in Figure 9. During the analysis, current density is kept constant at 0.8 A for first 600 s and then changed to 0.9 A for the remaining duration. These conditions gives rise to increase in cell temperature by about 6 °C with peak power reaching 400 mW cm⁻². The upshot of this cell temperature rise results in solid temperature to experience a huge jump of almost 10 °C. This temperature rise in the solid block if sustained then the cell temperature would rise further and its performance would start to deteriorate. The simulated dynamic response of this model

matches the experimentally obtained response for a voltage step from 0.7 V to 0.5 V with 0.1 V step size, [17]. A number of simulations and rigorous experimental investigations conclude that in order to maintain the cell within safe temperature limits, an efficient control strategy must be developed. Towards that end, in the following section we first develop a linearized version of this dynamic model and then design an explicit MPC.

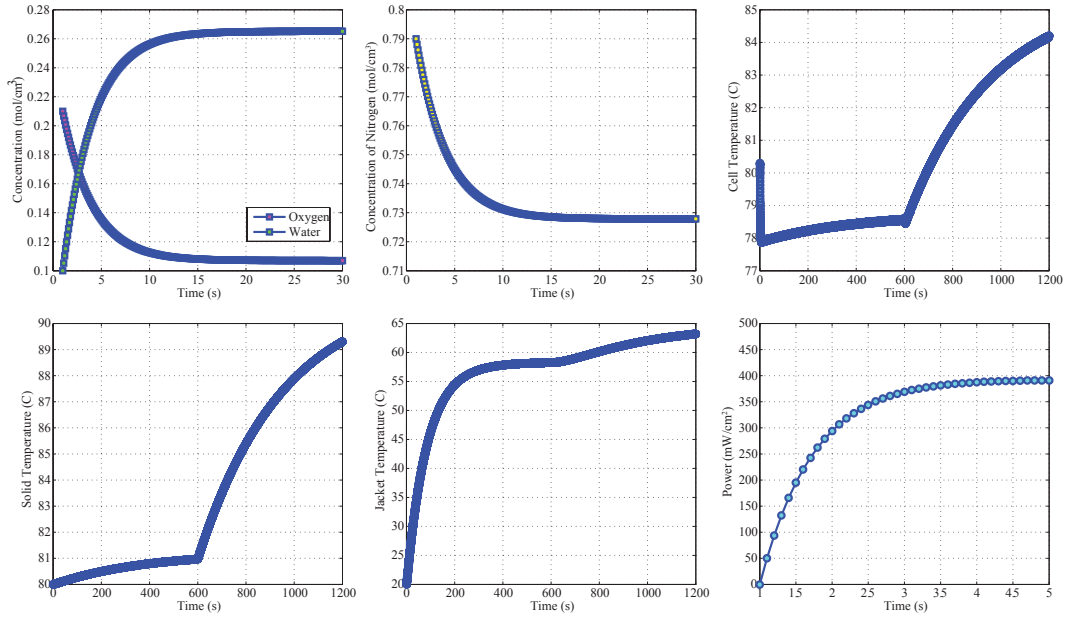


Figure 9: Dynamic response of system states

6. Model Approximation

For an appropriately chosen operating point or steady state setpoint, the first-order Taylor series expansion that results in the linearized dynamical system model in deviation variables (also known as state space model) is

given by,

$$\dot{x} = Ax + Bu + Gw \quad (29)$$

where vectors in deviation variables $x = [\tilde{C}_{O_2}, \tilde{C}_{H_2O}, \tilde{C}_{N_2}, \tilde{T}, \tilde{T}_s, \tilde{T}_j, \tilde{P}_e]$ represent states, $u = [\tilde{j}, \tilde{F}_c^{in}, \tilde{F}_j]$ represent inputs, $w = [\tilde{T}_c^{in}, \tilde{T}_j^{in}]$ represent known disturbances and constant A , B , G represent state, input and disturbance transition matrices. The quantities in deviation variables are given by the deviation of actual variable from its steady state or operating point, e.g., $\tilde{T} = T - \bar{T}$. The transition matrices are given by the first order derivatives of the nonlinear dynamics evaluated at the steady state or operating point as follows,

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial C_{O_2}} & \frac{\partial f_1}{\partial C_{H_2O}} & \frac{\partial f_1}{\partial C_{N_2}} & 0 & 0 & 0 & 0 \\ \frac{\partial f_2}{\partial C_{O_2}} & \frac{\partial f_2}{\partial C_{H_2O}} & \frac{\partial f_2}{\partial C_{N_2}} & 0 & 0 & 0 & 0 \\ \frac{\partial f_3}{\partial C_{O_2}} & \frac{\partial f_3}{\partial C_{H_2O}} & \frac{\partial f_3}{\partial C_{N_2}} & 0 & 0 & 0 & 0 \\ \frac{\partial f_4}{\partial C_{O_2}} & \frac{\partial f_4}{\partial C_{H_2O}} & \frac{\partial f_4}{\partial C_{N_2}} & \frac{\partial f_4}{\partial T} & \frac{\partial f_4}{\partial T_s} & 0 & \frac{\partial f_4}{\partial P_e} \\ 0 & 0 & 0 & \frac{\partial f_5}{\partial T} & \frac{\partial f_5}{\partial T_s} & \frac{\partial f_5}{\partial T_j} & 0 \\ 0 & 0 & 0 & 0 & \frac{\partial f_6}{\partial T_s} & \frac{\partial f_6}{\partial T_j} & 0 \\ \frac{\partial f_7}{\partial C_{O_2}} & \frac{\partial f_7}{\partial C_{H_2O}} & 0 & \frac{\partial f_7}{\partial T} & 0 & 0 & \frac{\partial f_7}{\partial P_e} \end{bmatrix},$$

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial j} & \frac{\partial f_1}{\partial F_c^{in}} & 0 \\ \frac{\partial f_2}{\partial j} & \frac{\partial f_2}{\partial F_c^{in}} & 0 \\ \frac{\partial f_3}{\partial j} & \frac{\partial f_3}{\partial F_c^{in}} & 0 \\ \frac{\partial f_4}{\partial j} & \frac{\partial f_4}{\partial F_c^{in}} & 0 \\ \frac{\partial f_5}{\partial j} & 0 & 0 \\ 0 & 0 & \frac{\partial f_6}{\partial F_j} \\ \frac{\partial f_7}{\partial j} & 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{\partial f_4}{\partial T_c^{in}} & 0 \\ 0 & 0 \\ 0 & \frac{\partial f_6}{\partial T_j^{in}} \\ 0 & 0 \end{bmatrix}$$

where functions $f_1, f_2, f_3, f_4, f_5, f_6, f_7$ are the chosen dynamics of interest given by equations (14, 15, 16, 18, 19, 20, 26) respectively. Based on the experimental observations, [17], we pick an operating conditions so as to maintain the operation in safe operating region. Evaluating the partial derivatives at oxygen concentration 0.21, water concentration 0.1, nitrogen concentration of 0.78, cell temperature 80 °C, solid temperature 80 °C, jacket temperature 60 °C, power density 0.5 mW cm⁻², and current density 0.83 mA cm⁻² results in the following continuous-time system matrices,

$$A = \begin{bmatrix} -0.27967 & 0.07434 & 0.07434 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 0.00354 & -0.35047 & 0.00354 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 0.27613 & 0.27613 & -0.07788 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 125.01912 & 125.01912 & 125.01912 & -1.07257 & 0.71828 & 0.00000 & -0.00108 \\ 0.00000 & 0.00000 & 0.00000 & -0.00094 & -0.00061 & 0.00020 & 0.00000 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.36069 & -0.36512 & 0.00000 \\ 300.69811 & -6314.66038 & 0.00000 & 8.34583 & 0.00000 & 0.00000 & -1.33333 \end{bmatrix} \quad (30)$$

$$B = \begin{bmatrix} 0.0000027 & -18.6252772 & 0.0000000 \\ 0.0000069 & 78.9356984 & 0.0000000 \\ -0.0000027 & -60.3104213 & 0.0000000 \\ 0.0001319 & -31321.5077605 & 0.0000000 \\ 0.0000272 & 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0000000 & -17738.3592018 \\ 0.6022516 & 0.0000000 & 0.0000000 \end{bmatrix} \quad (31)$$

$$G = \begin{bmatrix} 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0000000 \\ 0.0003333 & 0.0000000 \\ 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0044346 \\ 0.0000000 & 0.0000000 \end{bmatrix} \quad (32)$$

Application of zero order hold discretization method to the continuous

time state space model in equation (29) gives the following discrete-time linear time invariant state space model.

$$x(k+1) = A_d x(k) + B_d u(k) + G_d w(k) \quad (33)$$

where A_d is discrete-time state transition matrix, B_d is input transition matrix and G_d is disturbance matrix. This discretization of fuel cell systems with sampling time of one second gives the following system matrices,

$$A_d = \begin{bmatrix} 0.76447 & 0.06261 & 0.06261 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 0.00298 & 0.70485 & 0.00298 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 0.23254 & 0.23254 & 0.93441 & 0.00000 & 0.00000 & 0.00000 & 0.00000 \\ 76.52567 & 77.98610 & 76.59205 & 0.34054 & 0.43989 & 0.00005 & -0.00033 \\ -0.04244 & -0.04302 & -0.04247 & -0.00058 & 0.99918 & 0.00017 & 0.00000 \\ -0.00503 & -0.00509 & -0.00504 & -0.00011 & 0.30205 & 0.69414 & 0.00000 \\ 378.54647 & -2578.33787 & 244.14264 & 2.50922 & 1.40234 & 0.00010 & 0.26229 \end{bmatrix} \quad (34)$$

$$B_d = \begin{bmatrix} 0.0000025 & -15.6867882 & 0.0000000 \\ 0.0000058 & 66.4821024 & 0.0000000 \\ -0.0000014 & -50.7953142 & 0.0000000 \\ 0.0002515 & -19147.0673911 & -0.3074925 \\ 0.0000271 & 10.6217493 & -1.6058061 \\ 0.0000043 & 1.2600960 & -14862.2016349 \\ 0.3206758 & -210190.8354044 & -0.5314451 \end{bmatrix} \quad (35)$$

$$G_d = \begin{bmatrix} 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0000000 \\ 0.0000000 & 0.0000000 \\ 0.0002043 & 0.0000001 \\ -0.0000001 & 0.0000004 \\ 0.0000000 & 0.0037156 \\ 0.0006508 & 0.0000001 \end{bmatrix} \quad (36)$$

7. Design of Model Predictive Control

In order to find out the optimal control policy, we solve the following constrained optimal control problem for the discrete time linear time invariant system:

$$\begin{aligned} \Gamma(x(0)) &= \min_{u(\cdot)} \left\{ x(N)^T P x(N) + \sum_{k=0}^{N-1} x(k)^T Q x(k) + u(k)^T R u(k) \right\} \quad (37) \\ \text{s.t. } x(k+1) &= A_d x(k) + B_d u(k) + G_d w(k); \quad x(0) = x_0 \\ x(k) &\in \mathcal{X} \subseteq \mathbb{R}^n \\ u(k) &\in \mathcal{U} \subseteq \mathbb{R}^m \\ w(k) &\in \mathcal{W} \subseteq \mathbb{R}^p \quad ; \quad \mathcal{W} \equiv \{w_{lb} \leq w(k) \leq w_{ub}\} \end{aligned}$$

where $x(k)$ are states with $x(0) = x_0$ being the vector of initial states, $u(k)$ are control inputs, $w(k)$ are external disturbances. $Q \succeq 0$ and $R \succ 0$ are the weighting matrices for state and control while positive definite P is the stabilizing terminal cost for the prediction horizon N , which satisfies a certain Lyapunov criterion and is used to guarantee asymptotic stability of the MPC. Additionally, we assume that both $\mathcal{X}$ and $\mathcal{U}$ are closed polyhedral sets containing the origin in their interiors. Using the current state information, $x(0)$, problem (37) is solved to derive the predicted control policy, $u(\cdot)$ while only the 1st input sequence is implemented to derive the new state predictions, $x(\cdot)$. Now, using the new current states the problem (37) is solved repetitively and the procedure is continued at each time step in a receding horizon manner to obtain the next forecast of control actions.

We reformulate problem (37) into a multiparametric quadratic program

for the constraint set $\mathcal{X}$, $\mathcal{U}$ and $\mathcal{W}$ as

$$\begin{aligned} \Gamma(x(0)) &= \min_{U, W \in \mathcal{W}} [\Phi(x(0), U, W)] \\ \text{s.t. } &g(x(0), U, W) \leq 0; \quad \forall W \in \mathcal{W}^N \end{aligned} \quad (38)$$

where the column vector $U \triangleq [u(0)^T, \dots, u(N-1)^T]$ is the optimization vector, the column vector $W \triangleq [w(0)^T, \dots, w(N-1)^T]$ is the expected disturbance vector. After substituting

$$x(k) = A_d^k x(0) + \sum_{j=0}^{k-1} \{A_d^j B_d u(k-1-j) + A_d^j G_d w(k-1-j)\} \quad (39)$$

into the problem (37) gives $\Phi(x(0), U, W)$ as the quadratic objective function. Similarly, the substitution gives $g_j(x(0), U, W)$ as all the system constraints written in explicit form with $j = 1, \dots, J$ where $J = nN$. The solution of this multiparametric program gives the optimal control laws presented in section below.

7.1. Fuel Cell Control Laws

For state, input and disturbances constrained as the box constraints given by oxygen concentration ± 0.1 , water concentration ± 0.1 , nitrogen concentration ± 0.25 , cell temperature $\pm 40^\circ\text{C}$, solid and jacket temperature $\pm 10^\circ\text{C}$, current density $\pm 0.1 \text{ mA cm}^{-2}$ and disturbance variation of 1%, the optimal control law is given by the equation (40) and (41). Figures 10 and 11 show the evolution of states with this control feedback. Finally, figures 12, 13 and 14 show the time response of temperature of solid, temperature of the cooling jacket and the power generated. These results show that the open loop response and the unconstrained closed loop linear quadratic regulator is not

able to keep the system in given constrained bounds. On the other hand, the closed loop MPC is able to maintain the system in the operating limits.

$$\begin{aligned}
 U &= \begin{bmatrix} -99.7378 \\ -90.8629 \\ -99.3343 \\ -0.0024 \\ -0.0274 \\ 0.0000 \\ -0.0012 \end{bmatrix}^T \begin{bmatrix} c_{O_2} \\ c_{H_2O} \\ c_{N_2} \\ T \\ T_s \\ T_j \\ P \end{bmatrix} \\
 \text{if } & \begin{bmatrix} -0.1322 & 0.9879 & -0.0813 & -0.0010 & -0.0005 & 0.0000 & -0.0001 \\ -0.2255 & -0.1769 & -0.2233 & 0.0002 & -0.4808 & -0.7980 & 0.0000 \\ 0.1322 & -0.9879 & 0.0813 & 0.0010 & 0.0005 & 0.0000 & 0.0001 \\ 0.2255 & 0.1769 & 0.2233 & -0.0002 & 0.4808 & 0.7980 & 0.0000 \\ -0.1205 & -0.6909 & -0.7129 & 0.0005 & 0.0003 & 0.0000 & 0.0001 \\ -0.1403 & -0.9740 & -0.1782 & -0.0004 & -0.0010 & 0.0000 & 0.0001 \\ 0.0273 & 0.1732 & 0.0339 & 0.0004 & -0.9839 & -0.0001 & 0.0000 \\ 0.1205 & 0.6909 & 0.7129 & -0.0005 & -0.0003 & 0.0000 & -0.0001 \\ 0.1403 & 0.9740 & 0.1782 & 0.0004 & 0.0010 & 0.0000 & -0.0001 \\ -0.0273 & -0.1732 & -0.0339 & -0.0004 & 0.9839 & 0.0001 & 0.0000 \\ -1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{bmatrix} x + \begin{bmatrix} 0.0012 \\ 0.1709 \\ 0.0012 \\ 0.1709 \\ 0.2018 \\ 0.1245 \\ 9.8472 \\ 0.2018 \\ 0.1245 \\ 9.8472 \\ 0.1000 \\ 0.2500 \\ 40.0000 \\ 10.0000 \\ 200.0000 \\ 0.1000 \\ 0.2500 \\ 40.0000 \\ 10.0000 \\ 200.0000 \end{bmatrix} \quad (40)
 \end{aligned}$$

$$\begin{aligned}
U &= \begin{bmatrix} -200.3060 \\ 660.7800 \\ -161.1657 \\ -0.7301 \\ -0.4295 \\ 0.0000 \\ -0.0763 \end{bmatrix}^T \begin{bmatrix} c_{O_2} \\ c_{H_2O} \\ c_{N_2} \\ T \\ T_s \\ T_j \\ P \end{bmatrix} \\
\text{if } & \begin{bmatrix} -1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & -1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & -1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{bmatrix} x + \begin{bmatrix} 0.1000 \\ 0.1000 \\ 0.2500 \\ 40.0000 \\ 10.0000 \\ 10.0000 \\ 200.0000 \\ 0.1000 \\ 0.1000 \\ 0.2500 \\ 40.0000 \\ 10.0000 \\ 10.0000 \\ 200.0000 \end{bmatrix} \quad (41)
\end{aligned}$$

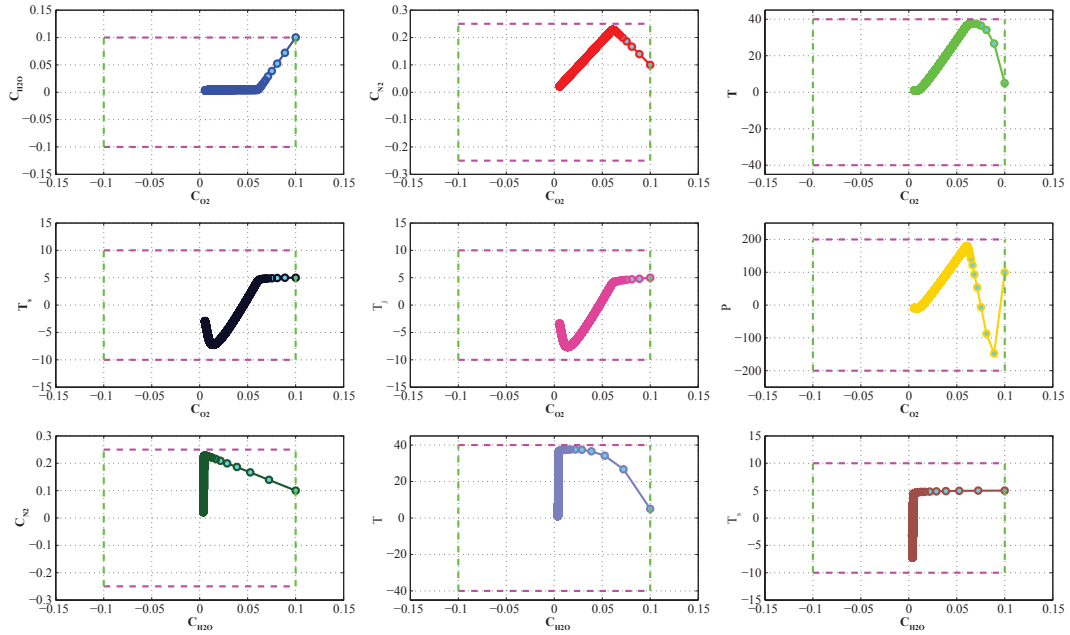


Figure 10: Evolution of states with control feedback

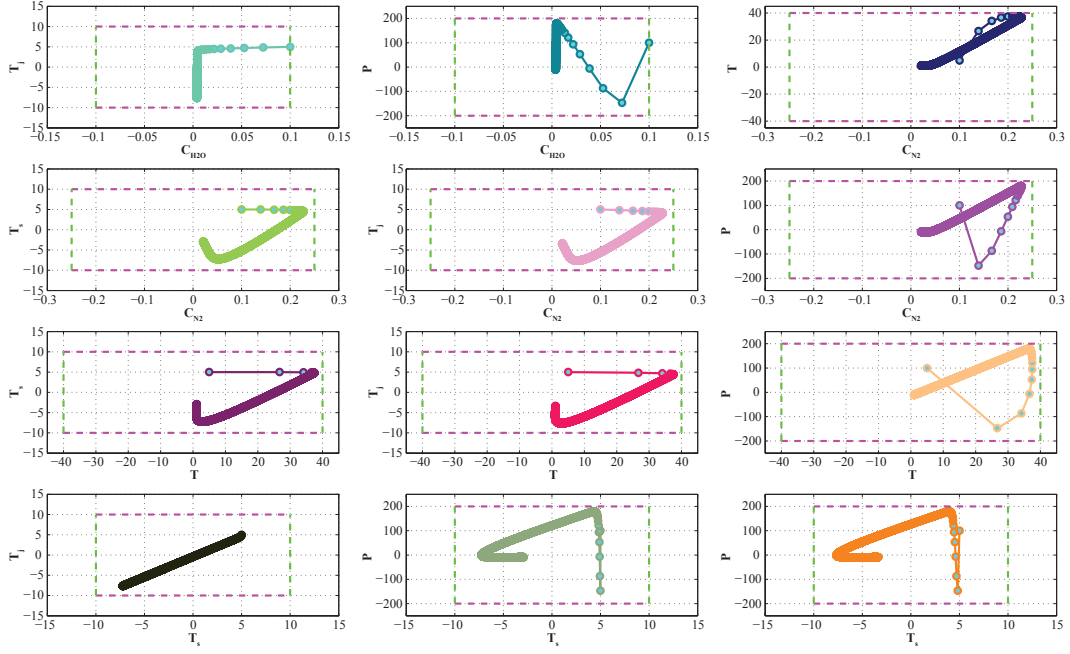


Figure 11: Evolution of states with control feedback

8. Conclusion

This work presented a framework for the development of efficient fuel cell controllers by integrating high fidelity mathematical models of the fuel cell system and integrated subsystems with online and offline model-based predictive control algorithms. A polymer electrolyte membrane fuel cell experimental platform acts as a testbed to demonstrate the capabilities of our smart control automation framework. The results demonstrate that the proposed embedded MPC-on-a-chip automation architecture is an ideal choice for operating fuel cell powered systems compared to using traditional PID type of regulatory controllers. The proposed control framework has an ability to easily interchange control algorithms and modelling innovations as soft-

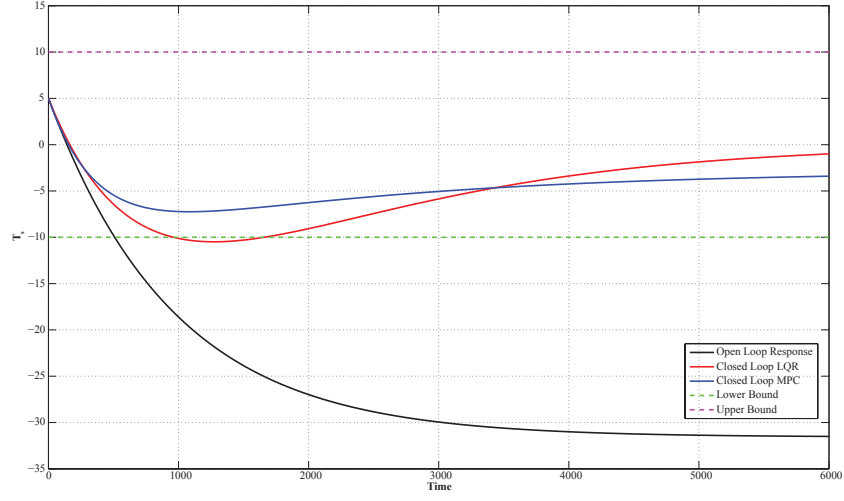


Figure 12: Time response of solid temperature with feedback control

ware functions. The proposed framework can further leverage the advances in cloud-based data architectures to achieve the ultimate goals of manufacturing intelligence.

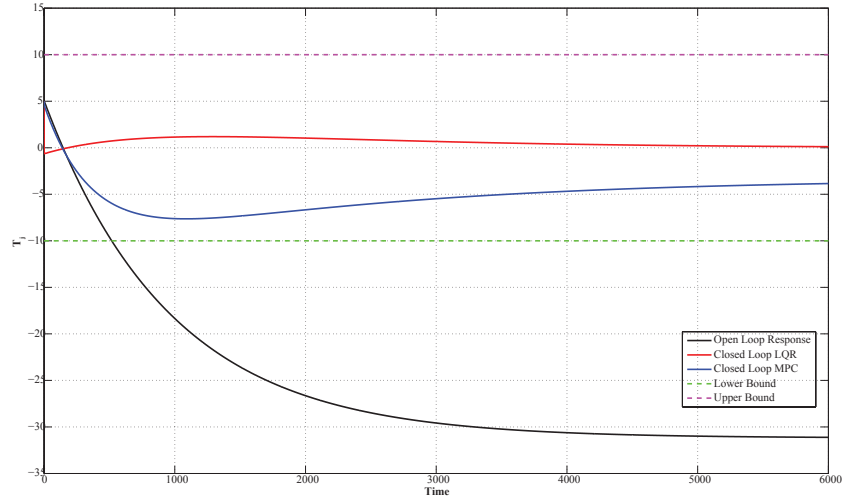


Figure 13: Time response of cooling jacket temperature with feedback control

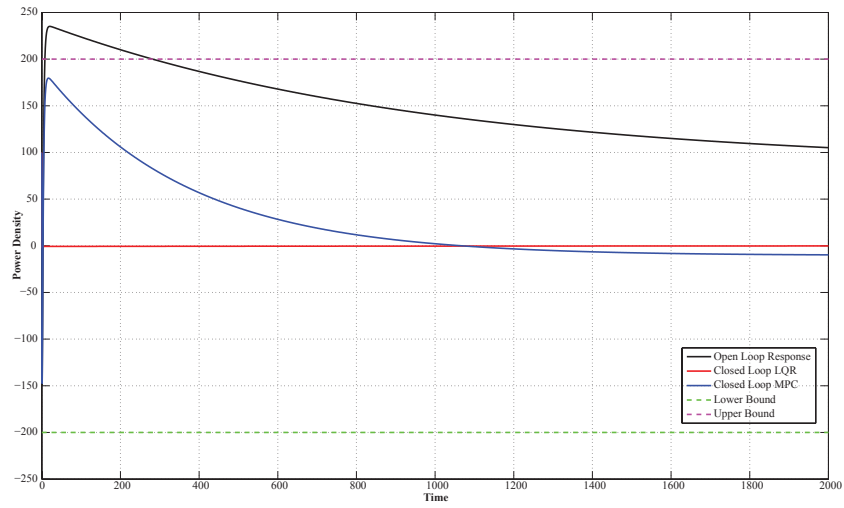


Figure 14: Time response of power generated with feedback control

9. Nomenclature

Symbol	Description
α	coefficient for activation overpotential
β	exponential factor for exchange current density
$\Delta H_{f_{H_2O(g)}}, \Delta H_{f_{H_2O(g)}}^\circ$	enthalpy of formation: water vapor, at STP
$\Delta G_{f_{H_2O(g)}}^\circ$	Gibbs free energy of formation of water vapor at STP
η, γ	cell: efficiency, resistance
ρ_a, ρ_s, ρ_j	molar density: air, solid, jacket
κ, ζ	coefficient: speed of reaction, concentration overpotential
σ, τ, θ	conductivity: ionic, exponent, exchange
ψ, ϕ	membrane moisture content, relative humidity
a_0, a_1, a_2, a_3	coefficients of moisture content
A_c, A_m, A_e, A_j	surface area: cathode, membrane, solid, jacket
$C_{H_2O}, C_{H_2O}^{in}$	concentration: water outlet, inlet
$C_{N_2}, C_{N_2}^{in}$	concentration: nitrogen outlet, inlet
$C_{O_2}, C_{O_2}^{in}$	concentration: oxygen outlet, inlet
$C_{p_a}, C_{p_{H_2}}, C_{p_{H_2O}}, C_{p_{O_2}}, C_{p_s}$	heat capacity: air, hydrogen, water, oxygen, solid
$E^\circ, E_{cell}, E_{ner}, E_{act}, E_{ohm}, E_{con}$	potential: open, cell, Nernst, activation, ohmic, concentration
F, n, R	constant: Faraday's, number of electrons, ideal gas
F_c^{in}, F_c^{out}, F_j	volumetric flow: cathode in, cathode out, jacket
j, j_L, j_{loss}, j_o	current density: cell, limiting, loss, exchange
P_e	Electrical power density
Q_e, Q_h	heat generated joules electrical, by reaction
r_{H_2O}	rate of reaction of water vapor formation
t_m	membrane thickness
$T, T^\circ, T_a, T_c^{in}, T_s, T_j, T_j^{in}$	temperature: cell, STP, ambient, cathode in, solid, jacket, jacket in
U_c, U_j, U_e	heat transfer coefficient: cathode, jacket, effective
V_c, V_s, V_j	volume: cathode chamber, solid, cooling jacket

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